

CMSE 801 - Day 2

Order of Magnitude Modeling

Sept 6, 2022

General Course Announcements

- Deadline for in-class assignments is at 11:59 p.m. the same day
- Deadline for pre-class assignments is at 5:30 p.m. either Tuesday for Wednesday's class or Friday for Monday's class
- Office hours:
 - Yann-Meing: Monday 3:00–5:00 p.m. at the Engineering Building room 2504
 - Jim: Friday 10:00 a.m. - 12:00 p.m. at the Engineering Building room 2504 and online
- Day-01 in-class assignment: algorithms

Questions/comments from Day 2 PCA

- I wasn't sure how many estimated ranges to iterate through for question 4 / Going over the last Question to make sure I actually understood it right and did it correct (which I feel like I didn't)
- Maybe I need more examples for above questions / I'd like to see more examples of order-of-magnitude modeling!
- As I mentioned in above, I struggled some with the language in the questions, but I think that better understanding will develop with time in this course
- Maybe if for the computer is the same saving a floating point than a string.
- How can people collaborate through the Jupyter notebook?
- the olympic pool question, and just the basic commands
- Would you give us the correct answer for each pre or in class assignments?
- Nothing much so far, just that my group originally thought we had to submit the exact same in-class assignment work on the first day, but I realized before submitting that we should each write the assignment in our own words (at least I think that is the case...)
- I have missed my first assignment as I was enrolled late in class. Let me know how I can redo the first missed assignment.
- So far its going great! / Nope! Once again I am very excited and grateful to be in this course / thanks :)

```
In [1]: # type of variable
a = 1
b = 2
print(type(a))
```

```
print(type(b))
print(a+b)
```

```
<class 'int'>
<class 'int'>
3
```

```
In [2]: c = 1.1
        print(type(c))
        d = b + c
        print(d)
        print(type(d))
```

```
<class 'float'>
3.1
<class 'float'>
```

```
In [3]: a1 = '1'
        b1 = '2'
        print(a1+b1)
        print(type(a1))
        print(a1+b)
```

```
12
<class 'str'>
```

```
-----
TypeError                                Traceback (most recent call last)
<ipython-input-3-63bb52743e1b> in <module>
      3 print(a1+b1)
      4 print(type(a1))
----> 5 print(a1+b)

TypeError: can only concatenate str (not "int") to str
```

```
In [4]: b2 = str(b)
        print(type(b2))
        print(a1+b2)
```

```
<class 'str'>
12
```

```
In [5]: firstName = 'Yann-Meing'
        lastName = 'Law'
        fullName = firstName + " " + lastName
        print(firstName)
        print(fullName)
        print(fullName[0])
```

```
Yann-Meing
Yann-Meing Law
Y
```

Order of Magnitude Modeling

How much time do I spend driving my car per day?

Assumptions

- only go to work
- only take highways

Inputs

- distance between work and home: l in km
- speed: v in km/h

Output

- time: t in hours

Ranges

- $l \in [100, 110]$
- $v \in [100, 120]$
- $t \in [t_{min}, t_{max}]$

Equations

$$t = 2 \frac{l}{v}$$

$$t_{max} = 2 \frac{l_{max}}{v_{min}}$$

$$t_{min} = 2 \frac{l_{min}}{v_{max}}$$

```
In [6]: lmin = 100
lmax = 110
vmin = 100
vmax = 120
tmin = 2*lmin/vmax
tmax = 2*lmax/vmin
print("t is between ", tmin, " hours and ", tmax, "hours." )
```

t is between 1.6666666666666667 hours and 2.2 hours.